After JIMWLK and BFKL: the remaining terms, the running coupling, DGLAP and the finite terms of your Group I, II, III

Companion of `Checks_JIMWLK_BFKL.pdf` (whose equations (I1)–(I5) are cited here). Your rows, without their log $\frac{k^+}{\Lambda}$ (JIMWLK) and log $\frac{\vee}{k^+}$ (BFKL) terms: blue = changed or added, red = removed or cancelling. Only $\vee > k^+$ and $p^+ > \Lambda$ are used; $\Lambda \to 0$ wherever possible, $\vee$ exact; every change is an exact identity. Groups I and III with c.c., Group II without.

1 What is done

1. **Remaining terms** (Secs. 2–4). Each row $= \log\frac{k^+}{\Lambda}\mathfrak{T} + \log\frac{\vee}{k^+}\mathfrak{P} + R$ exactly; here is R.
   - Every logarithm is replaced by its remainder (Sec. 3 of `Checks_JIMWLK_BFKL.pdf`), with $\Lambda \to 0$: $\left[\log\frac{k^+}{\Lambda} + \log(\frac{\vee}{k^+} - 1)\right] \to \log(1 - \frac{k^+}{\vee})$, $\log\frac{\vee}{\Lambda+k^+} \to 0$, $\log\frac{(\vee-k^+)A+k^+B}{\Lambda A+k^+B} \to \log\frac{(1-\frac{k^+}{\vee})A+\frac{k^+}{\vee}B}{B}$, …
   - Line 1 of Group III Rows III, V, VI and Group II Row IV is rewritten exactly, (I2)–(I5): what it contains of log $\frac{\vee}{k^+}$ is in the BFKL list.
   - $\Lambda \to 0$ everywhere else: $\mathcal{A} \to \mathcal{A}_0$; the limits of the plus-prescribed integrals ($\int_\Lambda^{k^+-\Lambda} \to \int_0^{k^+}$, $\frac{k^+}{k^++\Lambda} \to 1$); $\frac{1}{k^++\Lambda} \to \frac{1}{k^+}$; $2\log(1 - \frac{\Lambda}{k^+}) \to 0$.
   - Removed (red): the $\frac{1}{\Lambda}$ lines of Group I Row III (they add up to zero), your two red lines of d^3N_2 (they cancel), and the edge-kernel line of d^3N_{3b} (C4).

   Each of the 79 pointwise-evaluable terms equals its three parts to 1.5×10^{-14}, and the remainder has the $\Lambda \to 0$ limit written here (3.5×10^{-8} at $\Lambda = 10^{-9}$; `ll_checks.py split`). The rewriting of line 1 is checked in `bfkl_rows/exact_check.py` ($\le 3.7 \times 10^{-15}$). These terms stay finite for $\vee \to \infty$ and $\Lambda \to 0$, apart from the open finite part of the $\mathcal{D}$ region (Q3).
2. **Running coupling** (Sec. 5) and **DGLAP** (Sec. 6) are taken out of Group I Row II part 1, Group I Row V 2b UV and Group II Row IV, with the coupling counterterm of LO.
3. **The finite terms** are what is left (Sec. 7); Sec. 8 is the result.

Plus prescriptions. Yours, and two more (Eqs. (I2), (I2ξ)):

$$\int dp^+ \left[\frac{1}{p^+}\right]_+ f = \int dp^+ \frac{f(p^+) - f(0)}{p^+}, \qquad \int dp^+ \left[\frac{k^+}{p^+(p^+ + k^+)}\right]_+ f = \int dp^+ \frac{k^+\,[f(p^+) - f(0)]}{p^+(p^+ + k^+)},$$

$$\int \frac{d\xi}{\xi^2}\,\frac{\xi}{(1-\xi)_+}\,g = \int \frac{d\xi}{\xi(1-\xi)}\,[g(\xi) - g(1)], \qquad \int \frac{d\xi}{\xi^2}\,\frac{\xi^2}{(1-\xi)_+}\,g = \int \frac{d\xi}{1-\xi}\,[g(\xi) - g(1)].$$

$$\mathcal{A}_0 = \frac{1}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}} \left\{ \arctan\left[\frac{(x-z)^2 - (x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right] + \arctan\left[\frac{(x-w)\cdot(x-z)}{\sqrt{(x-w)^2\,(x-z)^2 - [(x-w)\cdot(x-z)]^2}}\right]\right\} \tag{1.1}$$

2 Remaining terms: Group I (with c.c.)

2.1 Row I

Only four ρ's: no remaining terms (`4rho.tex` has only $\left[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\right]$ terms).

2.2 Row II

Row II, first part. Your bracket without its $-2\log\frac{k^+}{\Lambda}$ (JIMWLK). $\log(1 - \frac{k^+}{\vee})\log\frac{\vee}{k^+}$ stays here: its coefficient depends on $\vee$, and $0 \le -\log(1 - \frac{k^+}{\vee})\log\frac{\vee}{k^+} \le \log^2 2$ for all $\vee > k^+$. The pole bracket times $\frac{11}{3}$ is the β_0 term (Sec. 5).

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{32\pi^4}\,\frac{1}{k^+}\int_{w,w',x,x',} e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon} - 2\gamma - \log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.$$

$$\left[\frac{11}{3}+\log\left(1-\frac{k^+}{\vee}\right)\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\Big]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^{b}(x)-U^{ab}(x)\,\rho^{b}(x)\Big] \tag{2.1}$$

Row II, second part, d^3N_{2a}. Nothing is left: this part has only $\log\frac{k^+}{\Lambda}$ and $\log\frac{\vee}{k^+}$ terms.

Row II, second part, d^3N_{2b}. Unchanged: no large logarithm; R^i as in your file.

$$\frac{d^3N_{2\,b}}{d^3k}=-\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'}e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y}e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{hbc}\,U^{ah}(w)\left\{\rho^b(x),\rho^c(y)\right\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\Big] \tag{2.2}$$

2.3 Row III

Row III, d^3N_1. Blue: what is left of the logarithms after their log $\frac{k^+}{\Lambda}$, log $\frac{\vee}{k^+}$ parts are taken out, with $\Lambda\to 0$. Red: the $\frac{1}{\Lambda}$ lines (your green brackets). Lines 1 and 4 here, line 1 of d^3N_2 and line 1 of d^3N_{3a} add up to zero exactly, pointwise (3.0×10^{-16}).

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-z)^2+(x-w)^2]}{[(x-z)^2-(x-w)^2]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-w)^2+(x-z)^2]}{[(x-z)^2-(x-w)^2]^2}\times\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{[(x-w)^2+(x-z)^2]}{[(x-z)^2-(x-w)^2]^2}\times\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\left[\frac{\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\} \tag{2.3}$$

Row III, d^3N_2. As d^3N_1. Red: line 1 is a $\frac{1}{\Lambda}$ line; your two red lines cancel each other.

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}=-\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,[(x-w)^2-(x-z)^2]}\left[\frac{1}{(\vee-k^+)}-\frac{1}{\Lambda}\right]$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y)-U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$
\begin{aligned}
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{2.4}
$$

Row III, d^3N_{3a}. Red: line 1 is a $\frac{1}{\Lambda}$ line (see d^3N_1). The edge-kernel line has no remaining part: $\log\frac{k^+-\Lambda}{\Lambda}=\log\frac{k^+}{\Lambda}+\log(1-\frac{\Lambda}{k^+})$. $\mathcal{A}_0$ is $\mathcal{A}$ at $\Lambda=0$, Eq. (1.1).

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{\\
&\frac{ig^4 f^{abd}}{8\pi^5}\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\\
&+\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}_0\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2}
\end{aligned}
$$

$$
\begin{aligned}
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0 \int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0 \int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned}
\tag{2.5}
$$

Row III, d^3N_{3b}**.** $\Lambda \to 0$. **C4**: the last line (red) is, for $\Lambda \to 0$, $\log\frac{(z-w)^2}{(x-w)^2}$ times the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$: not integrable at $z \to w$. Its logarithmic partner was replaced by $\mathcal{D}$ (F3); its finite part belongs to the $\mathcal{D}$ region (open, Q3).

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\mathcal{A}_0 \int_{x,y,z,x'} \frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right] \int_{x,y,z,x'} \frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&+ \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right] \int_{x,y,z,x'} \frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \\
&- \frac{ig^4 f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{(z-w)^2}{(x-w)^2}\right] \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big] \Bigg\}
\end{aligned}
\tag{2.6}
$$

2.4 Row IV

Row IV. Blue: the remainders of the two logarithms.

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)} \Bigg\{ & \\
&\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2 + \frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \times \left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\
&\left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\
&+ \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2} \times \log\left(1-\frac{k^+}{\vee}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} + \frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right] \Bigg\} \times \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]
\end{aligned}
\tag{2.7}
$$

2.5 Row V

Row V, first part. Unchanged.

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
&\left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+ \delta_{jm}\,\delta_{ik}\right] \times \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \\
&\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big] \\
&- \frac{1}{(2\pi)^3}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\left[\frac{1}{q^+}\right]_+ \\
&\times \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big]
\end{aligned}
\tag{2.8}
$$

Row V, second part, d^3N_{2a}. Unchanged.

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{1}{[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{2.9}
$$

Row V, second part, $d^3N_{2b\,\mathrm{UV}}$. $2\log(1-\frac{\Lambda}{k^+})\to 0$; $-\frac{11}{6}$ is the β_0 term that cancels the $\frac{11}{3}$ of Row II part 1 (Sec. 5).

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[-\frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\\
&\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y).
\end{aligned}
\tag{2.10}
$$

Row V, second part, $d^3N_{2b\,\mathrm{finite}}$. $\Lambda\to 0$ in the limits of the plus-prescribed integrals (regular at the end points). In line 6, $2\log\frac{k^+-\Lambda}{\Lambda}\to 2\log\frac{k^+}{\Lambda}$ (JIMWLK) $+2\log(1-\frac{\Lambda}{k^+})\to 0$.

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+(k^+-p^+)}{k^{+2}}\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\,\left[\frac{k^+}{q^+}\right]_+\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
$$

$$
-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}\right]
$$
$$
\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y) \tag{2.11}
$$

3 Remaining terms: Group II (no c.c.)

3.1 Row I

Row I, first part, first line. Blue: remainders. Line 2 is entirely BFKL ($\log\frac{\vee}{\Lambda+k^+}=\log\frac{\vee}{k^+}-\log(1+\frac{\Lambda}{k^+})\to\log\frac{\vee}{k^+}$): nothing of it is left.

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]
\end{aligned}
\tag{3.1}
$$

Row I, first part, second line. As the first line.

$$
\begin{aligned}
\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{
&\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]\\
&+\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}\\
&\times\Big[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\Big]\times\Big[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\Big]
\end{aligned}
\tag{3.2}
$$

Row I, second part, d^3N_{2a}. $\Lambda\to 0$ in the second logarithm.

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\Bigg\{
&\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\times\\
&\Bigg[\frac{(x-w)\cdot(z-w')\,(x'-z)\cdot(z-w)}{(z-w)^2\,(z-w')^2}-\frac{(x-w)\cdot(x'-w')\,(z-w)\cdot(x'-z)}{2\,(z-w)^2\,(x'-w')^2}\\
&+\frac{(x-z)\cdot(x'-z)\,(x-w)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-w')^2}\\
&+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(z-w)\,(x-z)\cdot(x'-z)}{2\,(z-w)^2\,(x'-z)^2}\\
&-\frac{(x-z)\cdot(z-w')\,(x'-w')\cdot(x-w)}{2\,(x-w)^2\,(z-w')^2}-\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(x-z)}{4\,(x-w)^2\,(x'-z)^2}\Bigg]\times\\
&\left[\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{k^+}{\vee}\,(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]-\log\left[\frac{(x-w)^2}{(x'-w')^2}\right]\right]\Bigg\}\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Big]
\end{aligned}
\tag{3.3}
$$

Row I, second part, d^3N_{2b}. Blue: remainders.

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right. \\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times \\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right. \\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2\,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]}\times \\
&\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right. \\
&\left.+\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \qquad (3.4)
\end{aligned}
$$

Row I, second part, d^3N_{2c}. Blue: remainders.

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{(x'-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right. \\
&\left.-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\right] \\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{(x-z)^2}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\left[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2}-\frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2}\right. \\
&\left.+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\right]\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right] \\
&\left[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\right.
\end{aligned}
$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Bigg] \tag{3.5}$$

Row I, second part, d^3N_{2d}. Blue: remainders.

$$\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} = &+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned} \tag{3.6}$$

Row I, second part, d^3N_{2e}. Blue: remainders.

$$\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)
\end{aligned}$$

$$- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \tag{3.7}$$

Row I, second part, d^3N_{2f}. Blue: remainders; $\frac{1}{k^+ + \Lambda} \to \frac{1}{k^+}$.

$$\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,[(x-z)^2-(x-w)^2]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\\
&\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&- f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\left[\frac{1}{\vee}-\frac{1}{k^+}\right]\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{1}{(z-w)^2\,(z-w')^2}\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&- f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{(x'-w')^2}\right]\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&- f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x) + N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned} \tag{3.8}$$

3.2 Row II

Only four ρ's: no remaining terms (`4rho.tex` has only $\left[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}\right]$ terms).

3.3 Row III

Only four ρ's: no remaining terms (`4rho.tex` has only $\left[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}\right]$ terms).

3.4 Row IV

Row IV. Line 1 is rewritten exactly (`Checks_JIMWLK_BFKL.pdf`, Sec. 3):

- $\frac{\xi}{(1-\xi)_+} \to \xi + \frac{\xi^2}{(1-\xi)_+}$, Eq. (I2$\xi$). The difference, $-g(1)\log\frac{\vee}{k^+}$, is (ii) of the BFKL list; it cancels (i), the $\log\frac{\vee}{k^+}$ of line 2.
- The first blue line subtracts the $\log(y'-y)^2$ term of $\frac{1-\xi}{\xi}$ at $y'=y$. After the y' integral that term is $\left[\log\frac{\vee}{k^+} - (1-\frac{k^+}{\vee})\right]H_4\big|_{\log\to 1}$, Eq. (I4); its $\log\frac{\vee}{k^+}$ part is the BFKL term $\log\frac{\vee}{k^+}H_4$.
- The second blue line is the rest, $+(1-\frac{k^+}{\vee})H_4\big|_{\log\to 1}$, which stays here.

Line 2: $\left[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\right] \to \log(1-\frac{k^+}{\vee})$. In line 3, $\frac{k^+}{k^+ + \Lambda} \to 1$. Lines 1, 2 with the two blue lines are the DGLAP term (Sec. 6).

$$\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = &\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\xi\,(1-\xi)+\frac{(1-\xi)}{\xi}+\frac{\xi}{(1-\xi)_+}\xi+\frac{\xi^2}{(1-\xi)_+}\right]\right]\times\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{g^2}{4\pi^2}\,\frac{N_c}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y)}{(x-y)^2\,(x'-y)^2}\left[-\frac{d_\perp}{2}\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)\right]\\
&\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\,\frac{1}{\xi}}\left[\frac{(1-\xi)}{\xi}\right]\right]\times\left[U^{e'c}(y)-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
\end{aligned}$$

$$
- \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{8\pi^5}\frac{1}{k^+}\left(1-\frac{k^+}{\vee}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}
$$

$$
\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
$$

$$
+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.
$$

$$
\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
$$

$$
+\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{(1-\xi)}{\xi}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}-\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2\,(x'-y')^2\,(y'-y)^2}\,d_\perp\right]\times
$$

$$
\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x) \tag{3.9}
$$

4 Remaining terms: Group III (with c.c.)

4.1 Row I

Row I. Blue: remainders.

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.
$$

$$
-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left.-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]
$$

$$
\left.-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\right\}\times
$$

$$
\left[U^{ab'}(x')\,U^{c'd'}(y)\,U^{c'b}(z)\,\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)\,U^{d'e'}(y)U^{ac'}(w')\,\rho^{c'}(x')\rho^{e'}(y)\right]\times
$$

$$
\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.1}
$$

4.2 Row II

Row II. Blue: remainders.

$$
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\left\{\right.
$$

$$
-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]\times\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right.
$$

$$
\left.-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2}\right]
$$

$$
\left.-\frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\right\}\times
$$

$$
\left[U^{c'd'}(y)\,U^{ab'}(x')\,U^{bc'}(z)\,\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\,\rho^{b}(y)\rho^{b'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right] \tag{4.2}
$$

4.3 Row III

Row III, first part. Line 1 is rewritten exactly (`Checks_JIMWLK_BFKL.pdf`, Sec. 3):

- $\left[\frac{1}{p^+}\right]_+\to\left[\frac{k^+}{p^+(p^++k^+)}\right]_+ + \frac{1}{p^++k^+}$, Eq. (I2). The difference, $-f(0)\log\frac{\vee}{k^+}$, is (ii) of the BFKL list; it cancels (i), the $\log\frac{\vee}{k^+}$ of line 2.
- In the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term, $G(y')\to G(y')-G(z)$ (blue line). The $G(z)$ part is (B), Eq. (I3), in the BFKL list.

Line 2: $\left[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\right]\to\log(1-\frac{k^+}{\vee})$.

$$
\frac{d^3N_{1\,\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}\,(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}
$$

$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+ \left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$+\ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{8\pi^4}\,\frac{1}{k^+}\int_0^{\vee - k^+}\frac{dp^+}{2\pi}\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left. -\ \frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right]$$
$$-\ \frac{1}{(2\pi)^3}\,\frac{ig^4\,f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\right] \tag{4.3}$$

Row III, second part. As the first part, with the $-\frac{p^+}{k^{+2}}$ term in place of $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$. The last blue line is the part of its $G(z)$ term that is not $\log\frac{\vee}{k^+}$, Eq. (I5). **C5**: the measure is $\int_{w,x',y',z,x}$.

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}}{(y'-z)^2\,(x'-y')^2}\,\frac{1}{[p^+\,(x-z)^2 + k^+\,(x-w)^2]}$$
$$\left[\frac{\delta_{k'm}\,d_\perp\,p^+}{2\,(z-w)^2\,(p^+ + k^+)^2}\,\left[(x-z)^2 - (x-w)^2\right] + \frac{p^+\,\delta_{k'm}}{k^+\,(p^+ + k^+)}\left[\frac{(x-z)\cdot(z-w)}{(z-w)^2} - \frac{(x-z)\cdot(x-w)}{2\,(x-w)^2}\right.\right.$$
$$\left.\left. -\ \frac{\left[(x-z)^2 - (x-w)^2\right]}{2\,(z-w)^2}\right] + \frac{\delta_{k'm}}{(p^+ + k^+)}\left[\frac{(x-w)\cdot(z-w)}{(z-w)^2} + \frac{(x-z)\cdot(x-w)}{2\,(x-z)^2} - \frac{\left[(x-z)^2 - (x-w)^2\right]}{2\,(z-w)^2}\right]\right.$$
$$-\ \left[\frac{1}{p^+}\right]_+\left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\left[\frac{(x-w)^{k'}\,(z-w)^m}{(z-w)^2} + \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-z)^2}\right] - \frac{p^+}{k^{+2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-}{2\,(x-w)}\right.$$
$$\left. -\ \frac{1}{k^+}\left[\frac{(x-z)^m\,(z-w)^{k'}}{(z-w)^2} - \frac{(x-z)^m\,(x-w)^{k'}}{2\,(x-w)^2} + \frac{(x-w)^m\,(z-w)^{k'}}{(z-w)^2} + \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-z)^2}\right]\right]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,w',y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m}{(y'-z)^2}\,\frac{1}{[p^+\,(x-z)^2 + k^+\,(x-w)^2]}$$
$$\left[-\frac{p^+}{k^{+\,2}}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times$$
$$\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left. -\ \frac{(x'-z)^{k'}}{(x'-z)^2}\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$+\ \frac{1}{(2\pi)^3}\,\frac{g^4}{4\pi^4}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)^m\,(x'-z)^{k'}}{(w'-z)^2\,(x'-z)^2}$$
$$\left[\frac{(x-w)^2}{(x-z)^2\,(p^+ + k^+)\,[p^+\,(x-z)^2 + k^+\,(x-w)^2]}\left[\frac{(x-z)^{k'}\,(z-w)^m}{(z-w)^2} - \frac{(x-z)^{k'}\,(x-w)^m}{2\,(x-w)^2}\right]\right]\times$$
$$\left[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right]$$
$$-\ \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.$$
$$\left. +\ \frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\left[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\right] \tag{4.4}$$

4.4 Row IV

Only four ρ's: no remaining terms (`4rho.tex` has only $\left[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\right]$ terms).

4.5 Row V

Row V. As Row III.

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{ce}(x)\,U^{ad}(y)\,\rho^e(x)\rho^d(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^e(x)\rho^d(y)\right] \tag{4.5}$$

4.6 Row VI

Row VI. As Row III.

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(x'-y')^{k'}\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[\frac{\delta_{k'm}\,\delta_{ij}}{(p^+ + k^+)} - \left[\frac{1}{p^+}\right]_+\left(\left[\frac{k^+}{p^+\,(p^+ + k^+)}\right]_+ + \frac{1}{(p^+ + k^+)}\right)\delta_{jm}\,\delta_{ik'} - \frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times$$
$$\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{4\pi^4}\frac{1}{k^+}\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\,\frac{(y'-z)^m\,(y-w)^i\,(x-z)^j}{(y'-z)^2\,(y-w)^2\,(x-z)^2}$$
$$\left[-\frac{1}{k^+}\,\delta_{im}\,\delta_{jk'}\right]\left[\frac{(x'-y')^{k'}}{(x'-y')^2}\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right.$$
$$\left.-\frac{(x'-z)^{k'}}{(x'-z)^2}\left[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\right]\times$$
$$\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(1-\frac{k^+}{\vee}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\left[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\right]\times\left[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^d(y)\,\rho^e(x) - U^{ad}(y)\,\rho^d(y)\,\rho^b(x)\right] \tag{4.6}$$

5 Running coupling

Notation. $a \equiv \frac{k^+}{\vee}$, $\beta_0 = \frac{11}{3}N_c$ $(n_f = 0)$, $d_\perp = 2-\epsilon$, $c_0 \equiv 2e^{-\gamma_E}$, $\ell_r \equiv \log\frac{r^2\mu^2}{c_0^2} = 2\gamma_E + \log\frac{r^2\mu^2}{4}$, $\hat{E}_r \equiv \frac{2}{\epsilon} + \ell_r$; $X = x-y$, $X' = x'-y'$, $\Delta = y'-y$. LO in your notation, and LO with a weight f:

$$\mathrm{LO}[f] \equiv \frac{1}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,f\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\left[U^{e'c}(y') - U^{e'c}(x')\right]\left[U^{ec}(y) - U^{ce}(x)\right]\rho^{e'}(x')\rho^e(x),$$

$\mathrm{LO} \equiv \mathrm{LO}[1]$; $\mathrm{LO}_\xi[f]$ is the same with $e^{-ik\cdot(y'-y)} \to e^{-ik\cdot(y'-y)\frac{1}{\xi}}$. $P^{(r)}_{gg,+}(\xi) = 2N_c\left[\frac{\xi}{(1-\xi)_+} + \frac{1-\xi}{\xi} + \xi(1-\xi)\right]$ with the plus prescription of $[0,1]$ restricted to $[a,1]$ $(\int_a^1 \frac{d\xi\,g(\xi)}{(1-\xi)_+} = \int_a^1 \frac{g(\xi)-g(1)}{1-\xi}d\xi + g(1)\log(1-a))$, $P_{gg} = P^{(r)}_{gg,+} + \frac{\beta_0}{2}\delta(1-\xi)$. Only three rows have poles or β_0: Group I Row II part 1, Group I Row V 2b UV, Group II Row IV (lines 1, 2); all other remaining terms are finite. Their conversion to units of LO was checked against your rows before (`finite_v2/final_checks.json`); the bookkeeping is `dglap_rc/checks.py`: β_0 sum

0

, poles $-\frac{\alpha_s}{2\pi}\left(-\frac{2}{\epsilon}\right)\left[P_{gg}\otimes\mathrm{LO} - N_c\log(1-a)\mathrm{LO}\right] = 0$, finite parts $-(\mathrm{CMW} + V_\vee + \mathrm{Q1} + \mathrm{DGLAP}) = 0$.

The β_0 terms. Two rows have β_0: the $\frac{11}{3}$ of Group I Row II part 1 and the $-\frac{11}{6}$ of Group I Row V 2b UV:

$$\frac{d^3N_1^{\beta_0}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\frac{11}{3}\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]\ +\text{c.c.}\ =\ \frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big], \tag{5.1}$$

$$\frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\frac{11}{6}\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\ \ +\text{c.c.}\ =\ -\frac{\alpha_s\beta_0}{4\pi}\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big]. \tag{5.2}$$

With $y\to x$, $z\to w$ in (5.2), the measure, the phase, the kernel and the bracket are those of (5.1), the colour factor changes sign, and $\frac{g^4N_c}{16\pi^4}\frac{11}{6}=\frac{g^4N_c}{32\pi^4}\frac{11}{3}$. So

$$\boxed{\frac{d^3N_1^{\beta_0}}{d^3k}+\frac{d^3N_{2b\,\mathrm{UV}}^{\beta_0}}{d^3k}=0} \tag{5.3}$$

for every a: pole, γ_E, μ and $\log(x-w)^2$ all cancel (`checks.json`). The virtual correction (Row II) and the real emission at short distance (Row V 2b UV) carry opposite β_0 terms, and no $\beta_0\log(x-w)^2$, i.e. no running of the coupling at the emission distance, is left.

The coupling of LO. LO is $\propto g^2$. In $\overline{MS}$, $g^2\to g^2(\mu_R)\big[1-\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\big]$ ($d_\perp=2-\epsilon$), which adds

$$\left.\frac{d^3N}{d^3k}\right|_{\mathrm{CT}} = -\frac{\alpha_s\beta_0}{4\pi}\frac{2}{\epsilon}\,\mathrm{LO} = \frac{\alpha_s}{2\pi}\Big(-\frac{2}{\epsilon}\Big)\frac{\beta_0}{2}\,\mathrm{LO}. \tag{5.4}$$

This pole is the $\frac{\beta_0}{2}\delta(1-\xi)$ part of the DGLAP pole (Sec. 6). After the coupling is renormalized at μ_R and the fragmentation function at μ_F (both $\overline{MS}$), the only β_0 term of the result is the renormalization-group term

$$\boxed{\left.\frac{d^3N}{d^3k}\right|_{\beta_0} = \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO}\Big|_{\alpha_s(\mu_R)}} \tag{5.5}$$

It makes the result independent of μ_R: $\frac{d}{d\log\mu_R^2}\mathrm{LO}|_{\alpha_s(\mu_R)}=-\frac{\alpha_s\beta_0}{4\pi}\mathrm{LO}$. It vanishes for $\mu_R=\mu_F$; then LO is taken with $\alpha_s(\mu_F)$.

The constant. The non-logarithmic part of the bracket of Row II part 1, $-\frac{67}{9}+\frac{\pi^2}{3}$, gives

$$\frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\Big[-\frac{67}{9}+\frac{\pi^2}{3}\Big]$$
$$\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\Big]+\text{c.c.}\ =\ \boxed{\frac{\alpha_s}{2\pi}\,N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO}} \tag{5.6}$$

$N_c(\frac{67}{9}-\frac{\pi^2}{3})=2K$ with $K=N_c(\frac{67}{18}-\frac{\pi^2}{6})$, the constant of the two-loop cusp anomalous dimension that the CMW scheme absorbs into the coupling. It is the finite companion of β_0 in this row.

6 DGLAP

All poles. The poles of the remaining terms (Secs. 2–4), in units of LO (`dglap_rc/checks.json`):

G1 Row II part 1 + c.c.	$\frac{\alpha_sN_c}{4\pi}\frac{4}{\epsilon}\big[\frac{11}{3}+\log(1-a)\big]\mathrm{LO}$
G1 Row V 2b UV + c.c.	$\frac{\alpha_sN_c}{2\pi}\frac{4}{\epsilon}\big[-\frac{11}{6}+2\log(1-\frac{\Lambda}{k^+})\big]\mathrm{LO}\quad(\log(1-\frac{\Lambda}{k^+})\to 0)$
G2 Row IV, lines 1, 2 (with the blue lines)	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\int_a^1\frac{d\xi}{\xi^2}P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi$
coupling counterterm (5.4)	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\frac{\beta_0}{2}\mathrm{LO}$
sum	$\frac{\alpha_s}{2\pi}\big(-\frac{2}{\epsilon}\big)\Big[\int_a^1\frac{d\xi}{\xi^2}P_{gg}(\xi)\,\mathrm{LO}_\xi-N_c\log(1-a)\,\mathrm{LO}\Big]$

The first term is the DGLAP pole: it is absorbed in $D_{h/g}(\zeta,\mu_F)$ ($\overline{MS}$), with the convolution $\int_{k^+/\vee}^1\frac{d\zeta}{\zeta^2}D_{h/g}(\zeta,\mu_F)\frac{d^3N_g}{d^3k}(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta})$. The second is not (end of this section).

The DGLAP term. Group II Row IV, lines 1 and 2 of the remaining terms with the two blue lines (Sec. 3, Row IV), with the pole removed (your bracket $\big[-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1\big]=-\hat{E}_\Delta+O(\epsilon)$ becomes $-\ell_\Delta$; the -1 cancels against $-\frac{d_\perp}{\epsilon}$), i.e. [your lines 1 and 2 without $\log\frac{k^+}{\Lambda}$] $-\log\frac{\vee}{k^+}H_4$ (the blue lines and $\xi+\frac{\xi^2}{(1-\xi)_+}$ add up to this exactly):

$$\frac{d^3N_{\mathrm{DGLAP}}}{d^3k} = \frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\int_{x,y,x',y'}\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-2\gamma_E-\log\left(\frac{(y'-y)^2\,\mu_F^2}{4}\right)\right]$$
$$\times\left[\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\,e^{-ik\cdot(y'-y)\frac{1}{\xi}}\left[\xi(1-\xi)+\frac{1-\xi}{\xi}\right]+\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{\frac{1}{\xi}e^{-ik\cdot(y'-y)\frac{1}{\xi}}-e^{-ik\cdot(y'-y)}}{1-\xi}\right.$$

$$
\begin{aligned}
&\quad + \log\left(1-\frac{k^+}{\vee}\right) e^{-ik\cdot(y'-y)}\Big] \\
&\times \Big[U^{e'c}(y') - U^{e'c}(x')\Big]\,[U^{ec}(y) - U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x) \\
&- \frac{1}{(2\pi)^3}\,\frac{g^4 N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{\vee}{k^+}\Big] \int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2(x'-z)^2} \\
&\times \Big[U^{e'c}(z) - U^{e'c}(x')\Big]\,[U^{ec}(z) - U^{ce}(x)]\,\rho^{e'}(x')\,\rho^{e}(x) \\
&= \boxed{-\frac{\alpha_s}{2\pi}\int_{k^+/\vee}^1 \frac{d\xi}{\xi^2}\,P^{(r)}_{gg,+}(\xi)\,\mathrm{LO}_\xi\big[\ell_\Delta(\mu_F)\big] - \log\frac{\vee}{k^+}\,H_4,} \qquad \ell_\Delta(\mu_F) = \log\frac{(y'-y)^2\mu_F^2}{c_0^2}.
\end{aligned}
\tag{6.1}
$$

- The bracket in ξ is the plus distribution $\frac{1}{2N_c}P^{(r)}_{gg,+}$ on $[\frac{k^+}{\vee},1]$; its last term is your $\log(1-\frac{k^+}{\vee})$ line.
- $\mu_F^2\frac{\partial}{\partial\mu_F^2}D_{h/g} = \frac{\alpha_s}{2\pi}P_{gg}\otimes D_{h/g}$ cancels the μ_F dependence of $\ell_\Delta(\mu_F)$ and of (5.5).
- The last two lines ($-\log\frac{\vee}{k^+}H_4$) remove the $\log\frac{\vee}{k^+}$ that the ξ integral contains, Eq. (I4) (it is in the BFKL terms); with them the DGLAP term is finite for $\vee\to\infty$.
- Line 3 of the row (linear polarization) is finite and not DGLAP; it is a finite term (Sec. 7).

The pole that is left (as before, Q1). The second term of the pole, with its finite partner from Row II part 1:

$$
\left.\frac{d^3N}{d^3k}\right|_{\mathrm{Q1}} = \frac{\alpha_s N_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,\mathrm{LO}\big[\hat E_X + \hat E_{X'}\big]. \tag{6.2}
$$

It is not DGLAP (it is not $P_{gg}\otimes$LO) and not running coupling. It comes from the limit $\xi\ge\frac{k^+}{\vee}$ in the virtual term of Row II part 1 and vanishes for $k^+\ll\vee$, i.e. at mid rapidity in the high-energy limit. In AABKL the virtual term has $\int_0^1 d\xi$ and there is no such term (arXiv:1411.2869).

7 The finite terms, step by step

What is left of every row after the evolution (`Checks_JIMWLK_BFKL.pdf`), the running coupling (Sec. 5) and DGLAP (Sec. 6):

1. **Group I Row II part 1**, Eq. (2.1). Its pole bracket times $\frac{11}{3}$: the β_0 term, cancelled by Row V 2b UV, Eq. (5.1)+(5.2)= 0. Its constant $-\frac{67}{9}+\frac{\pi^2}{3}$: Eq. (5.6). Its pole bracket times $\log(1-\frac{k^+}{\vee})$: Q1, Eq. (6.2). Left as a finite term:

$$
\frac{d^3N_{V\vee}}{d^3k} = -\frac{\alpha_s}{2\pi}N_c\log\Big(1-\frac{k^+}{\vee}\Big)\log\Big(\frac{\vee}{k^+}-1\Big)\,\mathrm{LO} \qquad \text{(with c.c.)}, \tag{7.1}
$$

   the sum of its $-\log^2(1-\frac{k^+}{\vee})$ and $-\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$ terms. It is bounded for every $\vee>k^+$ and vanishes for $\vee\to\infty$.

2. **Group I Row II part 2**: d^3N_{2a} has only $\log\frac{k^+}{\Lambda}$ and $\log\frac{\vee}{k^+}$ terms, nothing is left; d^3N_{2b}, Eq. (2.2), is finite and unchanged.

3. **Group I Row III**, Eqs. (2.3)–(2.6): finite (red lines removed: the $\frac{1}{\Lambda}$ lines add up to zero, your red lines cancel, C4).

4. **Group I Row IV**, Eq. (2.7); **Row V** parts 1, 2a, 2b finite, Eqs. (2.8), (2.9), (2.11): finite.

5. **Group I Row V 2b UV**, Eq. (2.10): nothing is left. $-\frac{11}{6}$ is the β_0 term (Sec. 5); $2\log(1-\frac{\Lambda}{k^+})\to 0$.

6. **Group II Row I**, Eqs. (3.1)–(3.8): finite.

7. **Group II Row IV**, Eq. (3.9): lines 1, 2 and the two blue lines are the DGLAP term, Eq. (6.1). Line 3 (linear polarization, $\frac{k^ik^j}{k^2}$) is left as a finite term.

8. **Group III Rows I, II, III, V, VI**, Eqs. (4.1)–(4.6): finite. In Rows III, V, VI the p^+ integrands of line 1 fall like $\frac{1}{p^{+2}}$; the rows are finite for $\vee\to\infty$.

All these terms are finite for $\epsilon\to 0$, $\Lambda\to 0$ and $\vee\to\infty$, apart from the open finite part of the $\mathcal{D}$ region (Q3). The terms with three ρ's among them need $\langle\rho\rho\rho\rangle$ (zero for a Gaussian projectile) or their commutator (two-ρ) parts.

8 Result

$$
\begin{aligned}
\frac{d^3N_h}{d^3k} &= \int_{k^+/\vee}^1 \frac{d\zeta}{\zeta^2}\, D_{h/g}(\zeta,\mu_F)\, \frac{d^3N_g}{d^3k}\Big(\frac{k^+}{\zeta},\frac{\mathbf{k}}{\zeta}\Big),\\
\frac{d^3N_g}{d^3k} &= \mathrm{LO}\big|_{\alpha_s(\mu_R)} + \frac{\alpha_s\beta_0}{4\pi}\log\frac{\mu_R^2}{\mu_F^2}\,\mathrm{LO} + \frac{\alpha_s}{2\pi}N_c\Big[\frac{67}{9}-\frac{\pi^2}{3}\Big]\mathrm{LO} \qquad \text{(running coupling, Sec. 5)}\\
&\quad + \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\,\mathrm{JIMWLK}\otimes\mathrm{LO} + \log\frac{\vee}{k^+}\,\mathrm{BFKL}\otimes\mathrm{LO}\Big] \qquad \text{(evolution, } \texttt{Checks_JIMWLK_BFKL.pdf}\text{)}\\
&\quad + \frac{d^3N_{\mathrm{DGLAP}}}{d^3k} \qquad \text{(Eq. (6.1))}\\
&\quad + \frac{\alpha_s N_c}{4\pi}\log\Big(1-\frac{k^+}{\vee}\Big)\,\mathrm{LO}\big[\hat{E}_X+\hat{E}_{X'}\big] \qquad \text{(Q1, Eq. (6.2))}\\
&\quad + \frac{d^3N_{V_\vee}}{d^3k} + \text{the finite terms of Sec. 7} + O(\alpha_s^3).
\end{aligned}
\tag{8.1}
$$

- $\mathcal{N}\big[\dots\big]$ is your JIMWLK formula (corrected, M2'; $\delta Y = \log\frac{k^+}{\Lambda}$) plus your BFKL formula ($\delta Y = \log\frac{\vee}{k^+}$), charge-averaged and with $\int_{w,w'} e^{-ik\cdot(w'-w)}$.
- Q1 and $V_\vee$ vanish for $\vee\to\infty$ at fixed k^+. Both come from the limit $\xi \geq \frac{k^+}{\vee}$ in the virtual term of Group I Row II; if the virtual term is taken with $\int_0^1 d\xi$ (as in AABKL, arXiv:1411.2869), both are absent.

9 Open points

Q1 The pole $\propto \log(1-\frac{k^+}{\vee})$, Eq. (6.2): neither DGLAP nor running coupling (see above).

Q2 A finite four-ρ term is not in your files (`4rho.tex` has only the $\big[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}\big]$ terms).

Q3 The finite part of the $\mathcal{D}$ region (the edge-kernel lines of Group I Row III, C4 and F3).